

WHEN A RECURRENT BREAST ABSCESS IS NOT JUST AN ABSCESS

Imaging clues to chronic granulomatous mastitis

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 BACKGROUND

- Chronic granulomatous mastitis (CGM) is a rare benign chronic inflammatory breast disease with an estimated incidence of 2.4 per 100 000 women.
- The clinical and imaging features frequently mimic abscess or breast carcinoma.
- Radiologists play a pivotal role in recognizing suspicious patterns and directing image-guided biopsy.
- Accurate tissue diagnosis guides appropriate multidisciplinary management and helps avoid unnecessary surgery.

OBJECTIVE / EDUCATIONAL AIM

- Highlight the evolving ultrasound appearance of CGM.
- Show how imaging redirected management from repeated abscess treatment to tissue diagnosis.
- Emphasise the importance of ultrasound-guided core biopsy of viable tissue.

CASE SUMMARY

- A 36-year-old non-lactating woman presented with a painful left breast lump that progressed to recurrent breast abscesses over a year despite repeated antibiotics and ultrasound-guided aspirations.
- Persistent suspicious imaging prompted ultrasound-guided core needle biopsy of the viable solid component.
- Histology confirmed chronic granulomatous mastitis.
- The patient was treated with corticosteroids, with gradual resolution on serial ultrasound.

Q WHEN SHOULD THE RADIOLOGIST SUSPECT CGM?

- ✓ Recurrent non-lactating mastitis/abscess.
- ✓ Poor response to antibiotics.
- ✓ Irregular hypoechoic inflammatory mass.
- ✓ Finger-like/tubular extensions ± multiloculated collections.

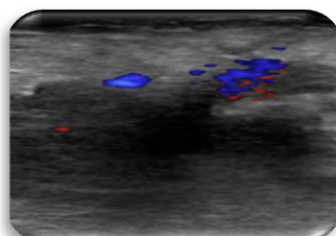
No single ultrasound feature is specific – interpret the imaging pattern together with the clinical course

"1" WHY ULTRASOUND MATTERS

- ✓ Characterizes evolving lesion morphology.
- ✓ Identifies suspicious viable solid components.
- ✓ Guides core biopsy while avoiding liquefied necrotic areas.
- ✓ Directs tissue diagnosis to exclude malignancy.
- ✓ Monitors treatment response and resolution.

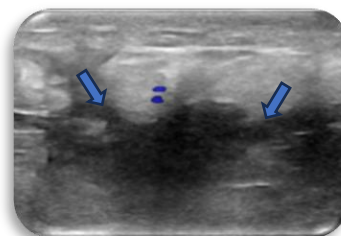
CHRONOLOGY OF BREAST ULTRASOUND FINDINGS

1 Initial presentation Month 0



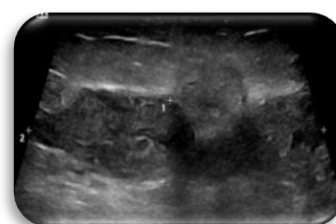
Irregular hypoechoic lesion with posterior acoustic shadowing, raising suspicion for malignancy.

2 Persistent / recurrent disease
After repeated antibiotics



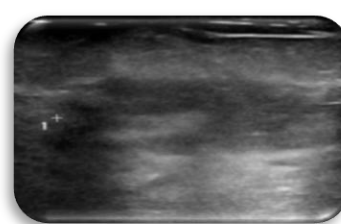
Finger-like tubular extension (blue arrow) tracking between breast parenchyma, with associated multiloculated abscess collection

3 Evolution of imaging pattern

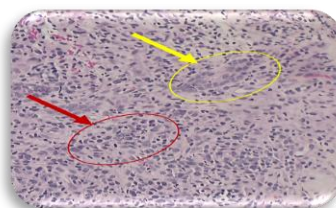


Persistent irregular inflammatory lesion despite antibiotics and repeated ultrasound-guided aspirations.

4 Follow-up / resolution
After corticosteroid therapy



Serial ultrasound demonstrated gradual reduction of inflammatory collections.

 HISTOPATHOLOGY CORRELATION

Core biopsy showed non-caseating granulomatous inflammation (red arrow) with multinucleated giant cells (yellow arrow) and mixed inflammatory infiltrates, without atypia or acid-fast bacilli – consistent with chronic granulomatous mastitis.

In recurrent 'breast abscess', recognising the evolving ultrasound pattern can redirect management from repeated drainage to tissue diagnosis – helping exclude malignancy and avoid unnecessary surgery.

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